

File Structure, Auto Importing of Data, X-TEC

HDF5 (.h5)

Acts as a virtual filesystem within one file

The tree like structure allows for big datasets to be managed through a single entry point

- Groups
 - The “folders” that create the tree
- Datasets
 - The “files” that hold actual numbers
- Attributes
 - Metadata

NeXus (.nxs)

- NeXus standardizes the name, locations of scientific metadata so automation software knows where to look
- NXentry, NXinstrument, NXsample, NXdata

Automatically Importing

How it runs:

1. Accepts a root directory
2. Finds and imports .nxs files
3. Sorts temperature folders numerically
4. Outputs a python dictionary
 - Packs all .nxs files into a master list
 - master_stacked_array

Using in other code:

- Data is formatted into standard grid
- Pass the variable or importing as a module

Challenge: Loading all 24 data files would use too much RAM

Solution: Uses lazy mapping, which means instructions are queued up but never actually executed unless a program asks to look at or calculate something

Testing With X-TEC

Challenges

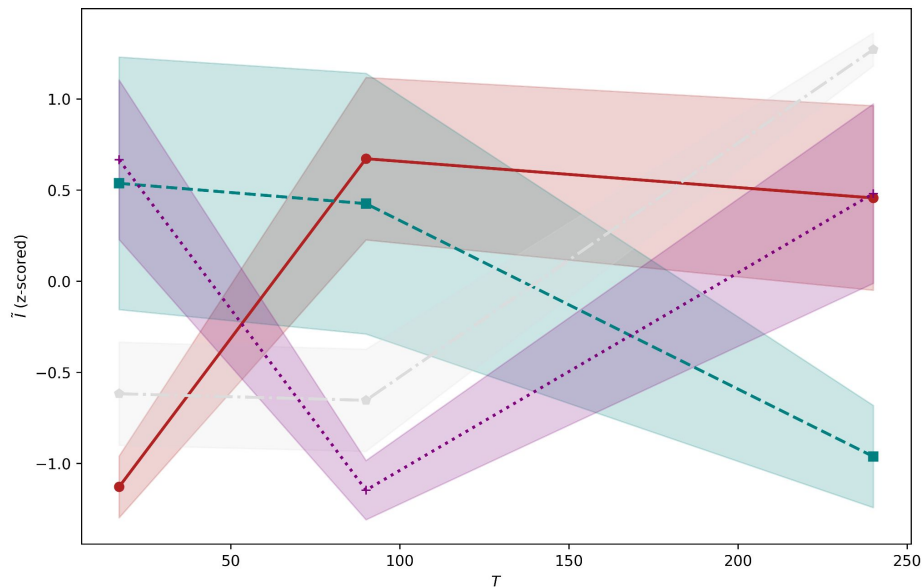
- Data structure mismatches
 - Wrong directory layout
 - Data and axes were in different files
 - HDF5 file names differed
- Going through all temperatures would take too long
- Hard coded output paths (pickles and figures)

Solutions

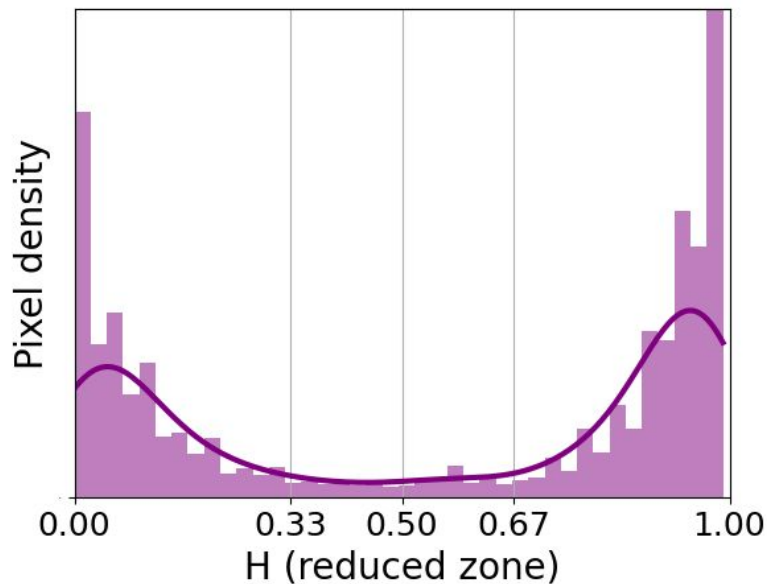
- Repointed data and folder structure
- Built two separate lists of file paths, one for each
- Used h5py to look at structure, find the data, and update
- Took a subset of three temperatures
- Replaced with plain file names so it would save to directory

Proof The Code Runs

These are not scientifically meaningful results since only three temperatures were used



Cluster Mean Intensity vs Temperature



One Brillouin Zone Histogram

Works Cited

1. The HDF Group. “HDF5 Data Model and File Structure.” *HDF5 User Guide*, 20 June 2026, support.hdfgroup.org/documentation/hdf5/latest/_h5_d_m__u_g.html.
2. Whittaker, Jeremy. “What Is an HDF5 File? A Deep Dive into Hierarchical Data Format.” *Medium*, 2 Nov. 2023, medium.com/@jeremywhittaker/what-is-an-hdf5-file-a-deep-dive-into-hierarchical-data-format-50327ec61731.